

Case Study Rubric
DS 4002- Spring 2026
Due: TBD

Individual Assignment

Why am I doing this? This case study allows you to use your data science knowledge and skills by applying text analysis techniques to a practical and customer-oriented issue for businesses: customer reviews of shipping companies. This assignment also asks that you practice sharing your work with relevant and tailored audiences. As you work through this assignment, you will gain hands-on experience in building machine learning models and interpreting their results to design solutions, exposing you to real-world applications of data analysis with relevancy in the business world.

What am I going to do? In this case study, you will work with a dataset of 120 customer reviews sourced from Trust Pilot, containing the customer review text as well as the date it was published and the star rating. You will take part in data collection for your clients, USPS, FedEx, UPS, and DHL, by scraping or inputting an additional 30 reviews per company. You are then tasked with exploring the dataset and developing a Natural Language Processing (NLP) and Latent Dirichlet Allocation (LDA) model using Python. Once the model is built, you will compare the results from each one to answer various questions and test hypotheses. The companies all assume they will have at least 50% of the comments be of negative sentiment and that most of the negative comments will be about delayed shipping because this is the industry average. You will test this hypothesis for each company and share the results. Then you will answer more questions about their performance outlined in the rubric table below under the “Results Brief”. Next, you will come up with recommendations for each company based on the results section and detail them in a “Recommendations Proposal”. Lastly, you will reflect on the project step by step and detail some of the limitations of your work, what you would do differently next time, and next steps for the case study.

Deliverables include:

- Updated dataset (csv) in “Data” folder
- Well-documented and functioning scripts utilized during case study (script notebook, py) in “Scripts” folder
- Any plots/tables generated (png, img, jpeg) in “Outputs” folder
- Results brief (pdf)
- Recommendations proposal (pdf)
- Written description of case study experience and reflections (pdf)
- GitHub repository containing all the above files

Link to provided GitHub repository (you will create your own):

<https://github.com/simoneminorr/Case-Study-Three/tree/main>

Tips for success:

- Do not simply think of this as a coding assignment; it is an application-based learning assignment where you will be asked to not simply submit your scripts, but the true implications of your work will be embodied by your brief.
- Collect the customer reviews across multiple days or when you need to do work but do not want to think critically instead of waiting until the last minute.
- Be effective, intentional, and creative. Think about what brief and proposal would be in alignment with your goal for this project. Given that you are acting as a data science consultant, it is probably to convince them to listen to you so considering using your generated visuals, aesthetic tables, and supporting research to make your points stronger.
- Use a notebook, like Jupyter or Google Colab for easier documentation.
- Notice, there are no accuracy metrics for you to reach on this project, because I want your attention to be focused on completing a well-rounded and applicable project. So, please put effort into the whole project, not just coding, but consider in your reflections how you could improve it perhaps in this section.
- Here are the links to the Trust Pilot reviews for each company:
 - USPS: <https://www.trustpilot.com/review/www.usps.com>
 - UPS: <https://www.trustpilot.com/review/www.ups.com>
 - FedEx: <https://www.trustpilot.com/review/www.fedex.com>
 - DHL: <https://www.trustpilot.com/review/www.dhl.com>

How will I know I have succeeded? You will meet expectations on this case study when you follow the criteria in the rubric below.

Formatting	• Create a GitHub repository for this assignment ○ Named as “DS4002_CS3_ShippingCustomerReviewsAnalysis” ○ Contains the following files: ▪ README.md ▪ LICENSE.md (use the MIT license) ▪ SCRIPTS directory ▪ OUTPUT directory ▪ Results Brief pdf ▪ Recommendation Proposal pdf. ▪ Reflections • Submit the link to this repository on Canvas
README.md	• <u>Goal:</u> A brief, high-level overview of your case study outcomes

	• It should contain: ○ 1-2 paragraph summary of the steps you took when reproducing the results, your overall experience doing the case study, and any suggestions for improvements ○ Link to the dataset (the one with your additions) ○ List of any external references you used (websites, journal articles, etc., cited in IEEE format)
SCRIPTS	• <u>Goal:</u> Contains all source code used in the case study • Upload the notebook used during the case study • Upload any scripts you have used with good and useful comments
OUTPUT	• <u>Goal:</u> Contains all plots/ tables generated • Uploaded any EDA (exploratory data analysis) plots that were generated • Upload a table containing model performance metrics • Use the provided output files as reference
Results Brief	• <u>Goal:</u> Describes the findings of the project. Opportunity to practice sharing results with a client. • Upload a report as a one-to-two-page pdf to your GitHub repository explaining your findings. ○ It should contain: ▪ Sentiment of customer reviews for each company ▪ Strengths and weaknesses for each company based on themes ▪ Answer the following questions: • The companies all assume they will have at least 50% of the comments be negative and that most of the negative comments will be about delayed shipping because this is the industry average. Please test this hypothesis for each company and share the results. • Who performs the best of the shipping companies? • Who performs the worst of the shipping companies? • What is the number one thing customers write negative reviews about? • What is the number one thing customers write positive reviews about?

	What is the number one thing customers write reviews about in general?
Recommendations Proposal	<u>Goal:</u> Make recommendations based on findings and comparisons for each company. - Upload a report as a one-to-two-page pdf with recommendations for each shipping company based on your findings to improve their company to your GitHub repository.
Reflections	<u>Goal:</u> Describe the limitations of your work and the next steps you would take. - Upload a report as a pdf (up to one page) to your GitHub repository with the following: - Limitations you faced going through each step of the process for this project - What you would do the same and differently next time - Next steps for this project

Acknowledgements: Special thanks to Professor Rasero and Rithwik Raman for their inspiration for this rubric.